

Variables and types



Agenda

- ① Variables
- ② Dynamic Type
- ③ type()
- ④ Data Types

Variables

- Variables are used to hold data during execution of the program.
- In C, C++, You need to declare variables. Only after declaration you can use them.

```
int a;  
float b;
```

C, C++ are statically typed languages

Variables

- In C, C++, You need to declare variables. Only after declaration you can use them.

```
int a;  
float b;
```

Python is dynamically typed language.

- In Python you don't declare variables. If there is a need of a variable you think of a name and start using it as a variable

C/C++

int a; ← declaration

a = 5;



a++;

printf("%d", a);

a = 7;

Python

a = 5 ← Automatic declaration

a = 5

a = a + 5

Variable name

- variable name is any combination of alphabet, digit and underscore.
- variable name cannot start with digit.
- variable names are case sensitive.
- Keywords cannot be used as variable names.

Deleting Variable

$x = 5$

del x

Dynamic Type

- Not only the value of a variable may change during program execution but the type as well.

$x = 5$ $\#$ type of x is int

$x = 5.7$ $\#$ type of x is float

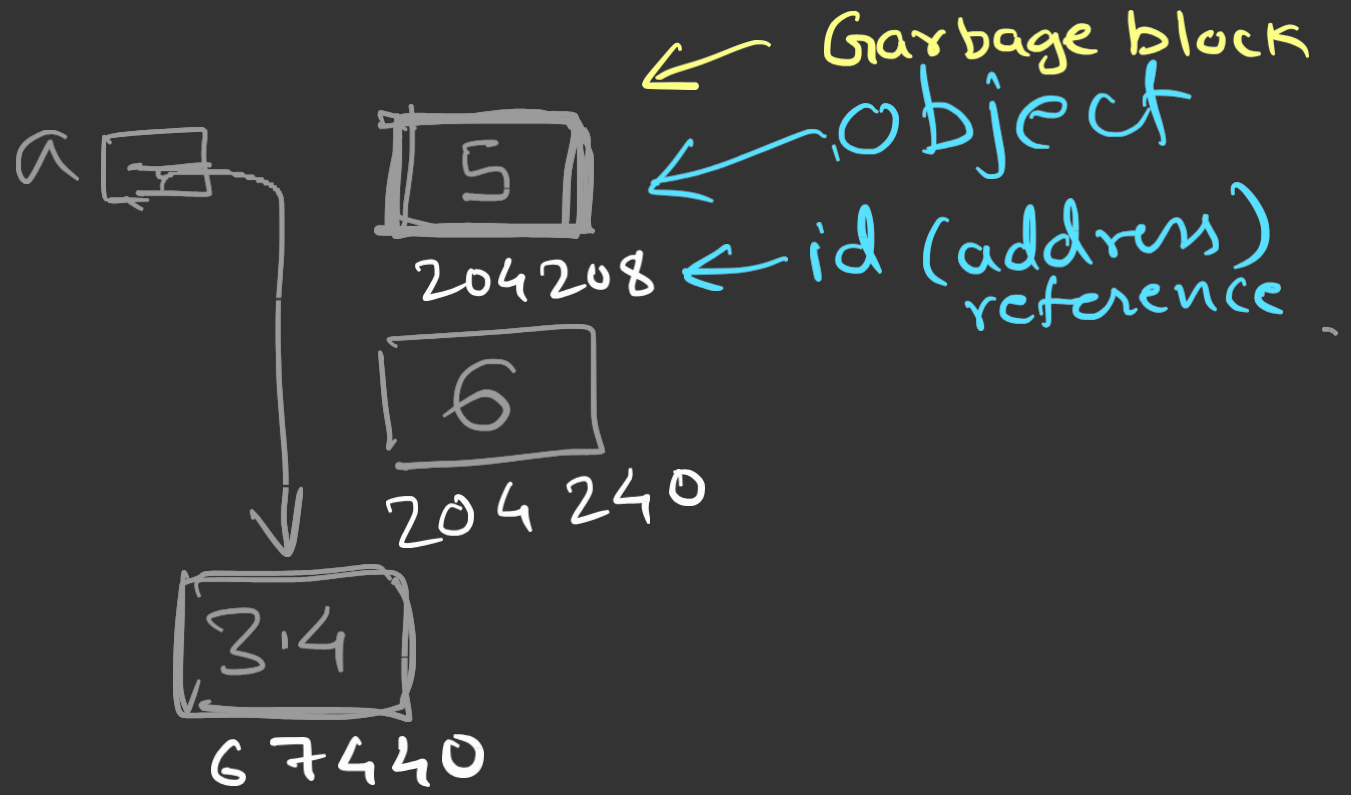
$x = \text{True}$ $\#$ type of x is bool

$x = \text{"MySirG"}$ $\#$ type of x is str

✓ a = 5

✓ a = 6

✓ a = 3.4



int
float
bool
str

data types
classes

C/C++
a
a = 5
a = 6

56

type()

- `type()` is a predefined function which returns the data type of a specified variable

```
x = 5
```

```
type(x)
```

```
x = 5.7
```

```
type(x)
```

data type is always a class in Python

Data Types

• Numbers

int	5
float	3.7
complex	3+4j

• Boolean

bool	→ True
	False

• String

str

"Tejas"
'Tejas'

"""Tejas"""
'''Tejas'''

double is not there in Python
char is not there in Python